Original Article

The Exploratory Use of HDPE Waste Plastic for Thermal Insulation Applications

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ABSTRACT

The study explored the possibility of replacing polyester polyol with HDPEPO in the development of rigid polyurethane foam as a thermal insulator. The HDPE waste plastics were prepared and pyrolysed at different temperatures (450, 475 and 500 °C) to yield high-density polyethylene pyrolysis oil (HDPEPO). The effect of different pyrolysis temperatures on important physicochemical parameters, such as density, yield, viscosity, iodine value and flash point, revealed that while density increases and flash point decreases with increasing pyrolysis temperature. Fourier-transform infrared spectroscopy (FTIR) and gas chromatography–mass spectrometry (GC–MS) analyses of the pyrolysis oil confirmed the presence of hydroxyl compounds in the HDPEPO, and their concentration decreased as the pyrolysis temperature increased, as higher temperatures promote the dehydration of alcohols to alkenes and water. The HDPEPO was used to formulate a polyurethane (PU) thermal insulator. The FTIR analysis carried out on the PU thermal insulator formulated revealed the presence of urethane linkage; however, it failed to rise, hence the need to modify HDPEPO with PEO to form HDPEPO/PEO blend. The effect of an increase in HDPEPO concentration on the crucial parameters (hydroxyl value, acid value, viscosity, and density) of HDPEPO/PEO blend revealed that the properties at 10% HDPE fall within the acceptable range. The characterisation of the HDPE-PU thermal insulator formulated using this blend is within the acceptable range when compared with the standard. The thermal insulator showed improvement in water absorptivity and thermal conductivity when compared with conventional thermal insulators. The percentage of HDPEPO utilised can be increased when PEO with a high hydroxyl value is deployed in the formation of HDPEPO/PEO blend.

Introduction

The plastics signifies a significant advancement in polymer technology, offering a combination of chemical resistance, mechanical strength, and environmental stability. According to [1], Hermann Staudinger discovered that a repeatable unit of polymer could resist chemical attack which was commonly occurs in other commodity materials such as metal, iron, and wood, thereby enjoying stability from stress and environment. Plastic production is cost-effective [2] and easy to produce [3], making it an economical replacement for conventional commodity materials such as iron, metals and wood. The plastic's unique properties such as lightweight, flexible, moisture resistant, strong, have contributed to its wide adoption across diverse spheres of life [4]. For instance, Gautam *et al.* reported that plastic finds applications in building and construction, textiles, consumer products and packaging,

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transportation, electrical and electronics, industrial machinery, and household commodities [5]. George [6] argued that plastics have revolutionized ways of life including crucial interactions with environment and technological integration.

The high-density polyethylene (HDPE) is a thermoplastic polymer that has become one of the most widely used plastics in modern manufacturing and construction applications. Its unique properties such as non-toxic, high impact strength, lightweight, and chemically inert, have enhanced its utility across many sectors. According to Labelnaterials [7], the global HDPE market size was valued at USD 17.96 billion in 2023 and is projected to reach USD 26.25 billion by 2030, exhibiting a CAGR of 5.57%. While HDPE plastic exhibits high patronage in water corporations, including water pipes and their fittings, however, much of its production has been centred on household commodities such as bottles (yogurt, some carbonated drinks, malts, chemicals, *etc*.); plates and cups (schools and homes), water buckets (different sizes), etc. The tremendous attraction to plastic, coupled with an undeniable behavioural propensity for increasingly over-consuming, discarding, littering and thus polluting, has contributed substantially to environmental pollution. Gill et al. reported an estimated 62% of the HDPE utilised for packaging applications [8].

The HDPE used for generating water and building fittings may take years to replace; unfortunately, the HDPE household commodity materials take few days especially the bottles after the content has been consumed, thereby generating huge waste for the environment. Moharir and Kumar argued that disposal of plastics is one of the least recognized and most highly problematic areas of plastic's ecological impact [9]. This is because discarded plastics after their use (waste plastic) never degrade and remain in the landscape for several years [10]. Gidigbi *et al.* [11] reported that the repeated unit of materials which prevented chemical attack is also responsible for nonbiodegradable of waste plastic. According to OECD, regrettably, 44.71 million tons of HDPE waste plastic were generated worldwide in 2019, a figure which may have doubled now due to increase in population [12]. The plastic waste is recyclable; however, it is not a permanent solution, as Hopewell *et al.* observed that only 70% of total waste may be worthy of recycling [13]. This is because recycling of a virgin plastic material can be done 2–3 times only, as after every recycling, the plastic material deteriorates due to thermal pressure resulting in a reduced lifespan [10].

Suvari and Gurvardar reported that recycled HDPE is mostly employed in the production of various items, including storage containers, boxes, pipes and toys [14]. This represents only few percentages of HDPE waste plastic, leaving amount that forms debris or buried within the soil. Other methods of taking care of handling HDPE waste plastics include incineration and landfills. Incineration leads to release of harmful gases into the atmosphere [10], while landfills responsible for groundwater poisoning [15].

Recent studies have focused on converting waste HDPE plastic to liquid fuel for energy consumption. For instance, Orozco *et al.* reported the fast pyrolysis of HDPE waste plastics using FCC spent catalysts to produce pyrolytic oil for energy consumption [16], while Panda *et al.* [10] reported the HDPEPO suitable to replace fossil fuel using zirconium hydroxide catalyst. Also, Suvari & Gurvardar studied the feasibility of converting waste HDPE to fibre yarn for textile industries [14]. While these processes prove to be efficient, however, the utilisation has been limited in some developing countries and rural communities due to lack of equipment (reactor) and technical know-how. There is a need for inclusive approach to tackle the increasing HDPE waste plastics. Thermal pyrolysis provides platform for an engagement at various levels. Its ease of fabrication at indigenous level and ease of operation are few of its many selling points. Other sources of heat energy such gas and briquette can be efficiently deployed. Thermal pyrolysis is temperature sensitive; thus, a need for process modulation at an appropriate temperature.

Polyol is one of the major reactants for the formulation of polyurethane thermal insulators. As a matter of fact, Prociak *et al*. [17] reported that polyol accounts for 30-50% of the total raw materials involved in polyurethane thermal insulator formulation. This means that substituting polyol with other renewable materials will lead to less dependence on petrochemicals and reduction

in cost of production. Therefore, this study assesses the feasibility of using HDPEPO for efficient thermal insulator in cooling systems.

Experimental Section

Materials

The materials used for this study included calcium carbonate, methylene chloride, magnesium sulphate, flame retardant, distilled water, methylene di-phenyl di-isocyanate (pMDI), polyester polyol, silicon oil, and stannous octanoate. All the reagents were of analytical grade. Pyrolyser, aluminum foil, weighing balance, infra-red spectrophotometer, stirrer, beakers, and 10 ml measuring cylinder, and stopwatch. The HDPE waste plastics were sourced around hostels and dump sites at Modibbo Adama University, Yola.

Pyrolysis of HDPE Plastics Waste

The HDPE waste plastic was identified (plastic code) and prepared by firstly washed with water and soap to eliminate oil and dirt. The plastic was dried at room temperature. The dried plastic was cut into pieces using scissors to ensure uniform heat treatment. The plastics pieces were introduced into a locally fabricated pyrolyser equipped with temperature controller. The pyrolysis temperature was raised to a selected temperatures (450, 475 , and 500 °C), while the pyrolysis oil was collected from the tap. Magnesium sulphfate was introduced into the pyrolytic oil to absorb possible moisture from the HDPEPO.

Determination of the Physicochemical parameter of the HDPEPO

The HDPEPO obtained at different temperatures (450, 475, and 500 °C) was subjected to physicochemical parameter assessment such as yield, density, iodine value, viscosity, and flash point. The suitability of each parameter for the HDPE-PU thermal insulation formulation was also discussed.

Determination of Chemical Constituents

The chemical constituents at different temperatures (450, 475, and 500 °C) were investigated using GC-MS (Agilent 5977C GC/MSD) and FTIR (Agilent Cary 640) using standard procedures.

Formulation of HDPE-PU Thermal Insulator using optimum HDPEPO

The formulation of rigid HDPE-polyurethane foam was carried out according to the method described by Członka *et al.* with slight modifications (Table 1). This method required two ways shot, where formulation was done in a reacting vessel before transferred to designated mold for casting [18].

Table 1: Chemicals used in the formulation of rigid HDPE-polyurethane thermal insulator

Components	Quantity (g)
HDPEPO	5.00
Silicone oil (Surfactant)	0.15
Distilled water (primary blowing agent)	0.20
Stannous octanoate (catalyst)	0.10
pMDI	6.00
Calcium Carbonate (Filler)	1.00

Optimisation of the HDPEPO/PEO Blend

The optimisation of HDPEPO/PEO was carried out by blending HDPEPO (5-20%) with polyester polyol to form HDPEPO/PEO blends. The effect of increasing in HDPEPO on the important parameters (such as acid value, hydroxyl value, viscosity, and density) of the HDPEPO/PEO blends was examined using the procedure outlined by Osemeahon *et al*. [19].

Characterisation of the Formulated HDPE-PU thermal Insulator and Comparison with the Standard

The rigid HDPE-polyurethane foam produced was subjected to different characterisations in order to ascertain its rigidity.

Results and Discussion

Pyrolysis of HDPE Plastic Waste

The HDPE waste plastics were successfully pyrolysed at different temperature (450, 475, and 500 °C) to give HDPEPO with a light-brownish colour and the char was dark brown. This is also in agreement with the findings of Marwani and Trifarizy, which also reported light-brownish colour for their HDPEPO [20]. The pyrolysis temperature was optimised to determine the suitable pyrolysis temperature (Figure 1).

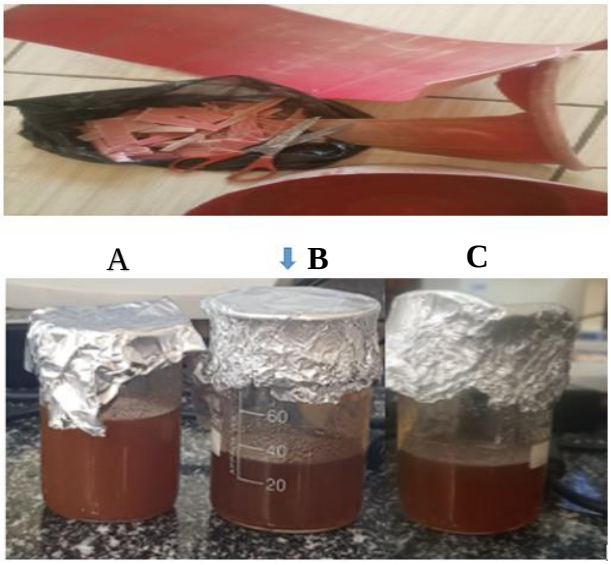


Figure 1: The HDPEPO at 450 °C (A), 475 °C (B), and 500 °C (C)

Physicochemical Properties of HDPEPO

Some of the important physicochemical parameters of the HDPEPO were examined under different temperatures (450, 475, and 500 °C) in order to achieve optimum reaction conditions for its usage.

Effect of Pyrolysis Temperature on the Yield of HDPEPO

Table 2 presents the effect of pyrolysis temperature on the yield of HDPEPO. The percentage yield of waste HDPE is similar to 12.02% yield of thermal pyrolysis of waste HDPE reported by Hendrawati *et al*. [21]. The percentage waste HDPEPO yield increases with an increase in temperature. This agrees with Prurapark *et al*. which reported that the yield of HDPEPO increases as the temperature increases [22].

However, Panda *et al*. [10] reported that the ideal pyrolysis temperature for thermal pyrolysis of most plastic wastes is to be within 450–700 °C, while Rominiyi *et al*. [23] reported that the maximum yield for HDPEPO using catalytic pyrolysis was observed to be 500 °C. The increase in temperature results in optimal yield due to the fact that waxes break to form liquid and char. As temperature of the pyrolysis increases, the breakdown of higher molecules increases which may lead to high yield of HDPEPO [24]. The highest yield of HDPEPO (500 °C) may be suggested for the rigid polyurethane thermal insulator formulation.

Table 2: Effect of pyrolysis temperature on the yield of HDPEPO

Temperature (°C)	Yield (%)
450	12.12±0.01
475	12.18±0.03
500	12.21±0.02

Effect of Pyrolysis Temperature on the Density of HDPEPO

According to Table 3, the density of the HDPEPO increases with increasing in pyrolysis temperature. This is due to the fact that aromatic compounds tend to evaporate as the temperature increases leaving aliphatic (longer chain of molecules) liquid with higher molecular weight [24]. A similar outcome was reported by Hendrawati *et al*. [21], which noted an increase in density of HDPEPO from 400 °C to 450 °C. The high-density polyol has a positive effect on the stability of the foam. Hence, HDPEPO at 500 °C is suggested for the formulation of HDPE-PU thermal insulator.

Table 3: Effect of pyrolysis temperature on the density of HDPEPO

Pyrolysis temperature (°C)	Density (g/cm^3)
450	0.671±0.02
475	0.673±0.01
500	0.674 ±0.02

Effect of Pyrolysis Temperature on the Viscosity of HDPEPO

Also, Table 4 revealed that the viscosity of the waste HDPEPO increases with increases in pyrolysis temperature. The viscosity behavior with temperature agrees with Panda *et al*. [10] which reported an increase in the viscosity of the pyrolysis oil as the pyrolysis temperature increases. The increase in viscosity as the temperature increases was a result of the increase in length of hydrocarbon chains [22]. Wang *et al*. reported that viscosity value of pyrolysis oil is comparable to conventional diesel fuel, thereby making it potentially suitable as a fuel substitute or blend component [25]. The viscosity of polyols affects their miscibility during thermal insulator production. Thus, HDPEPO at 475 °C may be suggested for rigid HDPE-PU thermal insulators.

Table 4: Effect of pyrolysis temperature on the viscosity of pyrolysis oil

Pyrolysis temperature (^{o}C)	Viscosity (mPa.s)
450	10.31±0.03
475	10.88±0.02
500	11.12±0.01

Effect of Pyrolysis Temperature on the Iodine Value of HDPEPO

Table 5 presents the effect of pyrolysis temperature on the iodine value of HDPEPO. The iodine value of HDPEPO increases as the pyrolysis temperature increases. This is due to the fact that thermal cracking favors unsaturated hydrocarbons as heavier hydrocarbons break down into smaller hydrocarbons, thereby, increases the unsaturated ring compound through secondary cracking and dehydrogenation reactions. The iodine value for HDPEPO in this study agrees with Uebe *et al*. who reported iodine value of 70 g/100 g for the HDPEPO [26]. Comparing the iodine value with biodiesel, Dagde reported a lower value of 38.2 g/100 g for avocado seed oil biodiesel [27]. However, a lower iodine value suggests a more saturated polyol, which may result in a more stable and durable thermal insulator. Thus, HDPEPO at 450 °C may be suggested for rigid HDPE-PU polyurethane formulation.

Table 5: Effect of pyrolysis temperature on the iodine value of the HDPEPO

Pyrolysis temperature (^{o}C)	Iodine value (g/100 g)
450	68±0.01
475	73±0.03
500	79±0.01

Effect of Pyrolysis Temperature on the Flashpoint of HDPEPO

According to Table 6, the flash point differs inversely with the fuel flammability. According to Gidigbi and Abubakar, the flash point is an indicator of lowest temperature at which oil can ignite. It measures the degree of flammability of the oil with respect to temperature. This may be due to the existence of lighter hydrocarbon fractions in the pyrolysis oil as the temperature increases.

Cahyono and Fenti observed similar trend of the decrease in flash point of a pyrolytic oil as the pyrolysis temperature increases [28]. This may be due to increases in lighter hydrocarbons in the pyrolysis oil as the temperature rises. In polyurethane production, flash point of an oil determines the safety of the production since the process is an exothermic in nature (releasing heat to the environment). Using oil with lower flash point may increase the chances of flammability during production. Therefore, HDPEPO at 450 °C may be suggested for the rigid HDPE-PU thermal insulator formulation.

Table 6: Effect of Pyrolysis Temperature on the flashpoint of HDPEPO

Pyrolysis temperature (°C)	Flash point (°C)
450	36.1±0.01
475	35.9±0.03
500	35.6±0.01

Comparative Study of the Chemical Compositions of HDPEPOs

Figure 2 presents the FTIR analysis of the HDPEPO at different temperatures of 450, 475, and 500 °C. The absorption observed at 3,280.40 cm^{-1} (450 °C HDPEPO); 3,274.90 cm^{-1} (475 °C HDPEPO) and 3272.10 cm^{-1} at 500 °C of HDPEPO suggestive of intramolecular alcohol.

The detection of hydroxyl group was also reported for HDPEPO via thermolysis [21]. Additionally, absorption at 3,080.6 cm^{-1} in all the HDPEPO's samples may be attributed to SP^2 of C-H bond of aromatic alkene. This is also supported by the absorption at 1,649.36 cm^{-1}, which affirmed the presence of olefin (vinylidene) compounds in the pyrolysis oil [27].

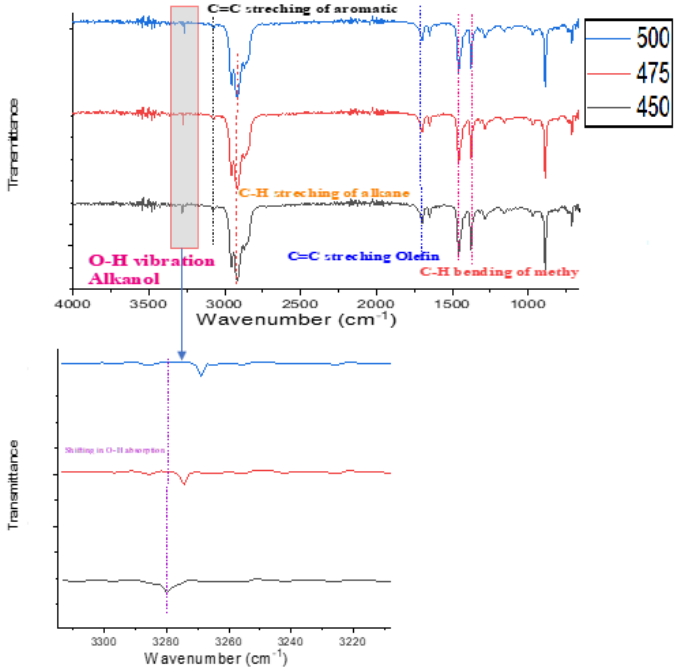


Figure 2: FTIR spectra of HDPEPO obtained at different pyrolysis temperatures

Moreover, the triplet peaks observed at 2,957.6, 2,911.1 and 2,871.9 cm^{-1} are attributed to sp3 of C-H bond of alkane. This was also supported by the absorption at 1,455.5 and 1,377.3 cm^{-1} signifying C-H bending of alkane [29,30]. The peak observed at 1,695.9 cm^{-1} signifies the presence of C=O of carboxylic/aldehyde. Additionally, the peaks observed at 1,151.7 and 969.1 cm^{-1} falls within the range of 1,225-950 cm^{-1} which suggestive of C-H out of plane bending for aromatic compounds [31]. This confirmed the existence of unsaturated compounds in the waste HDPEPO.

The Gas chromatography-mass Spectroscopy of HDPEPO

Table 7 presents the major chemical constituents of HDPEPO under selected temperatures of 450, 475, and 500 °C. The findings revealed the HDPEPO predominantly consists of straight-chain alkanes (C_8-C_{20}) and alkenes (1-alkenes), which is consistent with the thermal degradation mechanism of HDPE. The most abundant compounds include tridecane (9.58%), 8-chloro-1-octanol (8.37%), cyclohexanecarboxaldehyde (8.06%), and cyclooctane (7.90%). The light-hydrocarbon fraction of HDPEPO (C_4-C_8) is 32%, followed by medium hydrocarbon (C_9-C_{16}) at 30%, and heavy hydrocarbons (C_{17}-C_{20+}) accounting for 38% of the total composition. The predominant hydrocarbons (alkenes) in this study agree with the findings of Rodriguez Lamar *et al.* [32] and Sharuddin *et al.* [33], which reported similar hydrocarbon ranges in HDPEPO [20-22, 26, 27, 30-32].

Table 7: Major chemical constituents of HDPEPO at different temperatures

Major chemical constituents of HDPEPOs						Molecular formula
At 450 °C		At 475 °C		At 500 °C		
Area (%)	*Name of compound*	*Area (%)*	*Name of compound*	*Area (%)*	*Name of compound*	
9.58	Tridecane	9.01	Tridecane	8.81	Tridecane	$C_{13}H_{28}$
8.37	8-Chloro-1-octanol	8.21	8-Chloro-1-octanol	8.10	8-Chloro-1-octanol	$C_8H_{17}ClO$
8.06	Cyclohexane carboxaldehyde	7.89	Cyclohexane carboxaldehyde	7.78	Cyclohexane carboxaldehyde	$C_7H_{12}O$
7.90	Cyclooctane	7.87	Cyclooctane	7.82	Cyclooctane	C_8H_{16}
6.11	2-Butenoic acid	6.13	2-Butenoic acid	6.54	2-Butenoic acid	$C_4H_6O_2$
5.77	Cy clopentane	6.25	Cyclopentane	7.36	Cyclopentane	C_5H_{10}
4.67	Butyl tetratriacontyl ether	4.51	Butyl tetratriacontyl ether	4.47	Butyl tetratriacontyl ether	$C_{38}H_{78}O$
2.83	2-Hexyl-1-octanol	2.55	2-Hexyl-1-octanol	2.23	2-Hexyl-1-octanol	$C_{14}H_{30}O$
2.62	2-Methyl-E-7-octadecene	2.53	2-Methyl-E-7-octadecene	2.47	2-Methyl-E-7-octadecene	$C_{19}H_{38}$
2.51	1-Decanol	2.23	1-Decanol	2.10	1-Decanol	$C_{10}H_{22}O$
2.50	3-Hexene	4.01	3-Hexene	5.15	3-Hexene	C_6H_{12}
1.97	Dodecane	1.95	Dodecane	1.90	Dodecane	$C_{12}H_{26}$
1.76	8-heptadecene	1.51	8-heptadecene	1.43	8-heptadecene	$C_{17}H_{34}$
1.74	Cyclopentane	1.96	Cyclopentane	2.26	Cyclopentane	C_5H_{10}
1.49	1-Decanol	1.32	1-Decanol	1.24	1-Decanol	$C_{10}H_{22}O$
1.29	Henicos-1-ene	1.21	Henicos-1-ene	1.17	Henicos-1-ene	$C_{21}H_{42}$

The presence of hydroxyl compounds such as 8-chloro-1-octanol ($C_8H_{17}ClO$) and 1-decanol($C_{10}H_{22}O$) buttressed the findings of FTIR analysis of HDPEPO. This finding is consistent with Hendrawati *et al.* [21], which reported the presence of hydroxyl compounds

in thermal degradation of HDPE waste plastic using pyrolysis method. The oxygenated compounds are not part of the composition of HDPE; various factors such as HDPE additives have been suggested as sources of oxygenated compound in HDPE thermolysis process [33]. Moreover, comparing the hydrocarbons at the different temperatures (450, 475, and 500 °C), the hydrocarbon constituents C_{23}-C_{10} decrease as the pyrolysis temperature increases. This is due to the thermal breaking down of longer hydrocarbon chains, such as aliphatic chains, into smaller hydrocarbon chains. This phenomenon is supported by the study conducted by Grigore [34] on waste plastics recycling. Additionally, the slight decrease observed in hydroxyl groups as pyrolysis temperature increases may be connected to promotion of dehydration of alcohol to alkene and water as pyrolysis temperature rises.

Formulation of HDPE-polyurethane Thermal Insulator

The HDPEPO at 475 °C was chosen to substitute for conventional polyol and was completely used in the production of rigid HDPE-PU foam (thermal insulator). No cell formation or growth was observed. However, a whitish spot coalesced on the surface which later formed gel (Figure 3).

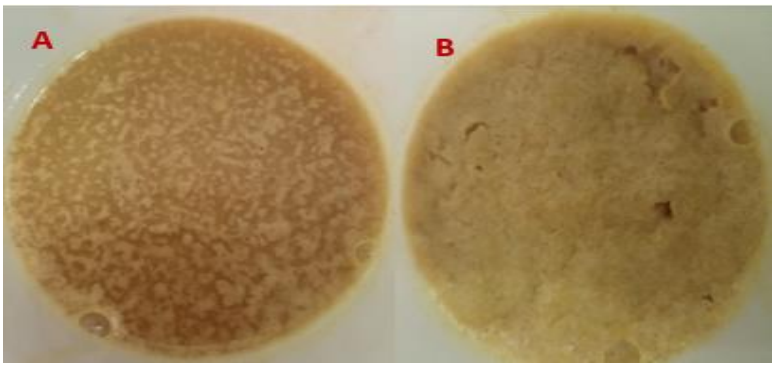


Figure 3: Rigid HDPE thermal insulator produced using HDPEPO (A) after production and (B) after one hour of production

Modification of the HDPE Pyrolysis Oil with Polyester Polyol

The modification of HDPE pyrolysis oil was carried out by blending 15% of HDPE pyrolysis oil with polyester polyol. The foam produced from this blend was dried and failed the basic test for rigid foam. The reason may be connected to the high acid value of the blend. Therefore, different blends of HDPE pyrolysis oil (5-20%) in polyester polyol were prepared according to the method described by Osemeahon *et al.* [19]. In order to avoid the error associated with foam formulated from the 15% HDPE pyrolysis oil in polyester polyol, the effect of increase in concentration of HDPE pyrolysis oil on important parameters of HDPE/PEO blends was investigated and the values were compared with the acceptable values for polyol that would be used in rigid foam formulation.

Effect of Increase in HDPEPO on the Acidic Value of the HDPEPO/PEO Blend

According to Turku *et al.* [35], the acid value for most conventional polyester polyols is within the 0.5-2.0 mg KOH/g range. The acid value of the HDPEPO/PEO blend (Figure 4) increases as the concentration of the HDPEPO increases. This is expected as the HDPEPO exhibits higher acidic value compared to polyester polyol. The blend exhibits dilution effect from the copolymers. Park *et al.* [36] observed that in a blend, the final acid value is not a simple weighted average of the components, as there can be chemical interactions between the HDPEPO and polyester polyol that affect the final acid value. However, Dutta [37] reported that in rigid PU thermal insulator formulations, lower acid values are preferred for better stability and reactivity.

Effect of Increase in HDPEPO on the Hydroxyl Value of the HDPEPO/PEO Blend

The hydroxyl value of the HDPEPO/PEO blend decreases as the concentration of HDPE pyrolysis increases (Figure 5). This may be due to the dilution effect of the HDPE polymer. The hydroxyl value in HDPEPO is very small, thereby diluting the concentration in the HDPEPO/PEO blend. Additionally, the decrease in hydroxyl value suggests that only the HDPEPO/PEO blend with 5-15% may be suitable for polyurethane thermal insulator production. The hydroxyl value of HDPEPO is generally relatively low, typically ranging between 15 and 45 mg KOH/g, reflecting the limited presence of hydroxyl groups in the pyrolysis oil products [36]. This characteristic is expected given that the original HDPE polymer contains no hydroxyl groups, and any hydroxyl groups present are primarily formed during secondary reactions in the pyrolysis process which may be attributed to additives such as ethylene glycol used in HDPE production.

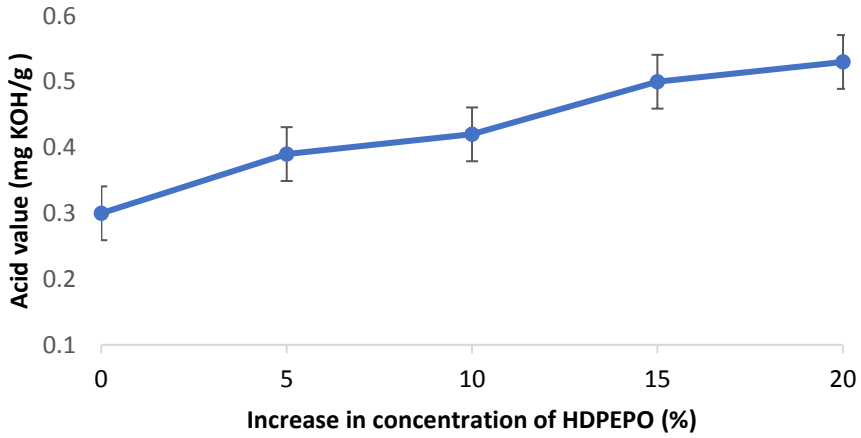


Figure 4: Effect of increase in the concentration of HDPEPO on the acid value of HDPEPO/PEO blend

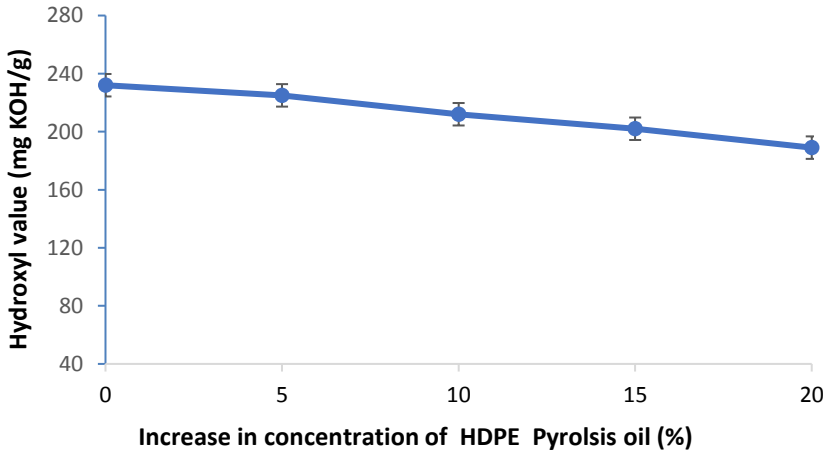


Figure 5: Effect of increasing HDPEPO concentration on the hydroxyl value of the HDPEPO/PEO blend

Effect of Increase in HDPEPO on the Viscosity of the HDPEPO/PEO Blend

Viscosity is primarily a rheological property of macromolecular polymers that demonstrates the barrier to flow. Figure 6 revealed the effect of increase in HDPEPO concentration on the viscosity of HDPEPO/PEO blend. The results exhibited a consistent decrease in viscosity of HDPEPO/PEO as the concentration of HDPEPO increases. This may be as an effect of the distinct chain length of the HDPEPO/PEO blend [38]. Osemeahon, *et al*. revealed that viscosity can be a means of investigating changes in density, colour, and molecular weight of a polymer [39].

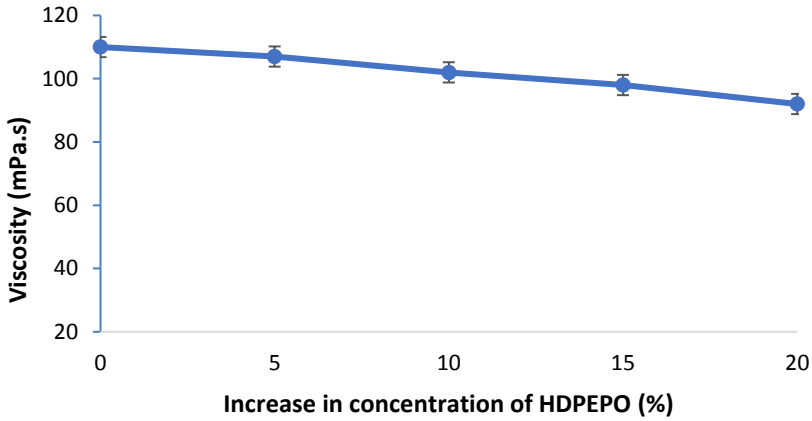


Figure 6: Effect of increase in HDPEPO on the viscosity of the HDPEPO/PEO blend

Effect of Increase in Pyrolysis Oil on the Density of the HDPEPO/PEO Blend

Density is a physical parameter of a substance, and is usually denoted in g/cm³. Density is an important criterion for determining the mechanical properties of a polymer. Figure 7 shows that the density of the HDPEPO/PEO blend decreases as the concentration of HDPEPO increases. Osemeahon *et al.* suggested that the observed decrease in the density of the HDPEPO/PEO blend may be as a result of an increase in the volume of the soft segment, as physical parameters of macromolecular materials depend on the chain length and content of the segments [40]. A similar trend was observed by Babanyaya *et al.* [38] when hydroxylated watermelon seed oil was copolymerised with polyvinyl acetate.

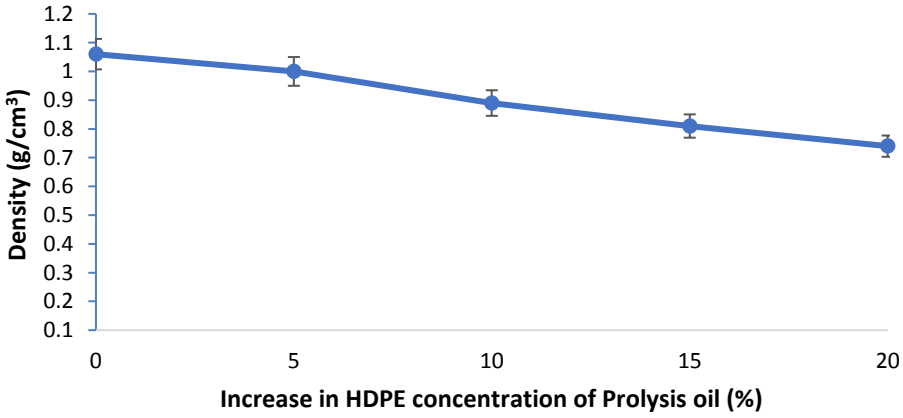


Figure 7: Effect of increasing HDPEPO concentration on the density of the HDPEPO/PEO blend

Comparison of Some Physical Properties of Different HDPEPO/PEO Blends with the Acceptable Value for Polyol Used in Rigid polyurethane Foam

Table 8 presents the physicochemical properties of different blends of HDPEPO in polyester polyol (HDPEPO/PEO). The important parameters such as acid value, hydroxyl value, and viscosity of 5% HDPEPO in polyester polyol were within the acceptable values for polyol. Meanwhile, these important parameters of 10% HDPE in polyester polyol were still within the boundaries of acceptable values for any polyol that will be used for production of polyurethane thermal insulators. However, at 15% HDPEPO in polyester polyol, the acid value of the blend outshoot exceeded the limit, whereas the hydroxyl value was at the edge of the minimum value required for polyol that is used in rigid foam formulation.

Table 8: Physicochemical parameters of different HDPEPO/PEO blends

Parameters	Percentage of HDPEPO in HDPEPO/PEO blends					Acceptable value for polyol in rigid foam production [41]
	0	5	10	15	20	
Acid value (mg KOH/g)	0.30±0.02	0.39±0.02	0.42±0.03	0.51±0.02	0.53±0.03	< 0.5
Hydroxyl value (mg KOH/g)	232±0.03	225±0.03	212±0.01	200±0.02	189±0.02	200–500
Viscosity at 25 °C (mPa.s)	110±0.03	107±0.02	102±0.01	98±0.02	92±0.03	< 50,000
Density (g/cm^3)	1.06±0.03	1.0±0.01	0.89±0.02	0.81±0.03	0.74±0.03	-

HDPEPO, HDPEPO/PEO Blend and the Acceptable Level for Polyol Used in Polyurethane Thermal Insulator Formulation

According to Ivdre *et al*. [42], unless a polyol meets some important physicochemical requirements, it might not be useful for rigid foam formulation. Thus, a need to compare of the HDPEPO/PEO blends with the acceptable values of polyol for rigid foam formulation. According to Table 9, a large disparity between polyester polyol and HDPEPO was observed in hydroxyl value, and viscosity. The acid value of HDPEPO is beyond the acceptable level for polyols. Additionally, the hydroxyl value for HDPEPO was too low for crosslinking to occur between the OH of HDPEPO and OCN of Methylene Diphenyl Diisocyanate (MDI). This may be the reason why foam formation failed when HDPEPO was completely substituted for polyester polyol in rigid foam formulation. This means that complete substitution of polyester polyol with HDPEPO might not be feasible for the formulation of rigid foam, even though, HDPEPO consists of hydroxyl groups. Furthermore, the physicochemical parameters of 10% blend of HDPEPO in polyester polyol fall within the acceptable range for polyols used in rigid foam formulation. Thus, 10% blend of HDPEPO in polyester polyol was adopted to form HDPEPO/PEO blend in which it was used to produce rigid foam (Figure 8).

Table 9: Physicochemical parameter of HDPEPO, polyester polyol, and HDPEPO/PEO blend

Physicochemical parameters	HDPEPO	Polyester Polyol	HDPEPO/PEO	Acceptable value for polyol in Rigid foam formulation [42]
Acid value (mg KOH/g)	0.57+ 0.03	0.30±0.02	0.42±0.03	< 0.5
Hydroxyl value (mg KOH/g)	30+ 0.02	232±0.03	212±0.01	200–500
Viscosity at 25 °C (mPa.s)	10.88+ 0.01	110±0.03	102±0.01	< 50,000
Density (g/cm^3)	0.673+ 0.03	1.06±0.03	0.89±0.02	-
Flash point (°C)	35.9	-	45.7	-

The experiments were done in triplicate with mean and standard deviation.



Figure 8: Production of rigid foam from HDPEPO/PEO blend

Characterization of the Formulated Rigid HDPE-PU Foam from HDPEPO/PEO

The absorption at 3,349.0 cm^{-1} suggestive of hydroxyl (O-H) group from alcohol. The absorption peak near 2,279.3 cm^{-1} was associated with unreacted NCO from excess pMDI. The dual absorptions observed at 2,972.6 and 2,868.2 cm^{-1} are associated with CH_3 and CH_2 chains of the polyol blend. The peak at 1,712 cm^{-1} suggestive of the C=O of carbonyl in polyurethane structure. The peak at 1,599 cm^{-1} was suggestive of CONH band while 1,541.3 cm^{-1} indicates the presence of aliphatic nitro compounds N-O which might be due to isocyanates [43]. The absorption at 1,658.7 cm^{-1} suggested of -C=C- from possible aromatic content of HDPEPO. The absorptions at 1,086.5 cm^{-1} and 1,455.5 cm^{-1} were indicative of C-O and C-N of polyurethane thereby confirming chemical interaction between HDPEPO/PEO and isocyanate. The peak at 1,508.6 cm^{-1} was observed, which was an indicative of C-H stretching of the aryl of isocyanate (Figure 9).

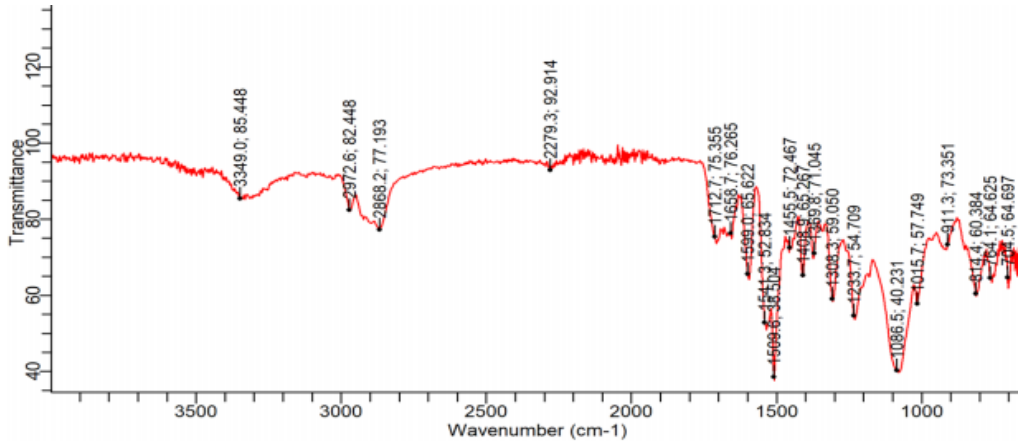


Figure 9: FTIR of the HDPE-PU foam

Scanning Electron Microscopy (SEM) Analysis

The SEM analysis in foam is solely conducted to determine the nature of the cell (open or closed cell) present in the polyurethane foam. The rigid foam exhibited more of closed-cell characterisitcs, while flexible foam exhibited more of open-cell properties. Figure 10 reveals the surface morphology of the rigid HDPE-PU foam made from the HDPEPO/PEO blend.

The microstructure of the HDPE foam revealed ccomplete cell wall formation, and cells appeared as a discrete, sealed units with clear separation from neighboring cells. The cell wall thickness and cell networking/crosslinking observed in the image are characteristic features that contribute to the foam's mechanical properties, as reported by Kairyte *et al.* [44]. There are few visible perforations or openings of cells. This slight perforation is also observed in production of polyurethane thermal insulators by Bello and Yan [23].

However, Fu *et al*. observed that rigid HDPE-polyurethane foams typically have smaller,

closed cells to improve thermal insulation and mechanical properties, whereas flexible foams have larger, open cells for compressibility and softness [45]. Also, the foam exhibits large pores which may be associated with inconsistent gas reactions during the urethane formation. Amaral *et al*. [46] reported that polyurethane foams with high porosity are often studied for their acoustic or thermal insulation properties.

Comparison of Physicochemical Properties of Rigid HDPE-Polyurethane Foam and Acceptable Value in Foam Industries

Table 10 presents the comparison of physicochemical properties of rigid HDPE-polyurethane foam produced from HDPEPO/PEO blend and the acceptable value for the polyurethane thermal insulator. The forming parameters such as cream time, gel time and tack-free time showed that they are within the generally acceptable range for polyurethane thermal insulators, with the exception of gel time. The low gel time was also observed by Suleiman *et al*. [47]. The low gel time may be attributed to moderate hydroxyl value of the HDPEPO/PEO blend, as the hydroxyl group available quickly cross-linked with isocyanate to form urethane linkages [42]. Also, gel time is also dependent on the quantity of catalyst used in the formulation. However, the density of the rigid HDPE-polyurethane foam produced fits into a generally acceptable range, while it failed to enter into the optimal value for structural applications. The reason may be attributed to the usage of surfactants which may affect the mixture of all the chemical components in the formulation.

Considering the water absorptivity of the HDPE-PU thermal insulator, the presence of HDPEPO in its polyol blend may contribute to low water absorptivity. This is also further proven that the foam exhibits more of closed-cell which is a typical instance of rigid foam. The result is close to the findings of Jabar *et al*. [48], who reported 0.95% for polyurethane thermal insulator formulated from vegetable polyols. The thermal conductivity of the rigid HDPE-polyurethane foam was evaluated to be 0.022 $W.m^{-1}\cdot K^{-1}$. This result agrees with the findings of Ivdre *et al.* [42], who reported 0.021 $W.m^{-1}\cdot K^{-1}$ for polyurethane thermal insulator made from RO/PET polyol. Additionally, Vieira *et al.* [49] reported 0.0401 $W.m^{-1}\cdot K^{-1}$ for thermal conductivity of polyurethane thermal insulator made from conventional polyol. The presence of hydrophobic HDPEPO in HDPEPO/PEO blend may be responsible for its good thermal behavior.

The compressive strength of the HDPE-polyurethane foam obtained from HDPEPO/PEO blend is 110 kPa. This value is lower when compared to the outcomes of Kirpluks *et al.* [41], who reported 350 kPa for rigid polyurethane foam formulated from bio-renewable polyol. Ivdre *et al.* [42] also reported 300 kPa for rigid polyurethane foam made from rapeseed oil/PET blend. Septevani *et al.* [50] reported that compression strength has direct relationship with the apparent density. The increase in density leads to increase in compressive strength of a foam. . The thermal conductivity of the rigid HDPE-polyurethane foam was within the acceptable level.

The closed cell content was determined by assessing SEM image of the foam cell as reported by Kirpluks *et al*. [41]. The HDPE-polyurethane foam cell observed to be closed with distinct cell walls [49].

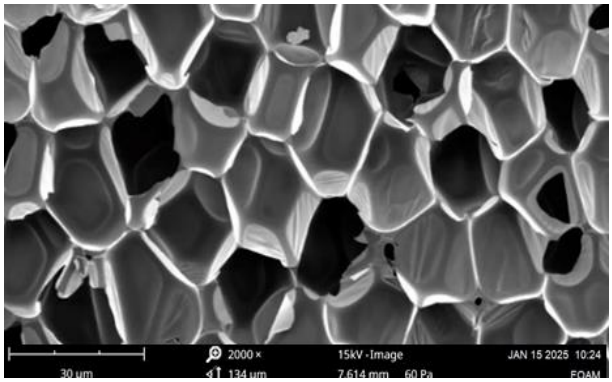


Figure 10: The SEM analysis of a rigid HDPE-PU foam formulated from HDPEPO/PEO polyol blend

Table 10: Comparison of the performance characteristics of HDPE-polyurethane foam with acceptable values for rigid foam

Parameters	Rigid HDPE-PU foam	Acceptable value [51,52]
Cream time (s)	9 ±0.02	5-15
Gel time(s)	17 ±0.03	30-120
Tack-free time (s)	63 ±0.01	30-120
Density (g/cm^3)	0.032 ±0.03	0.03-0.1
Water absorptivity (%)	1.1±0.04	<2
Compressive strength (kPa)	120 ±0.02	100-400
Thermal conductivity ($W.m^{-1}.K^{-1}$)	0.023 ±0.03	0.02-0.03

Conclusion

The study showed that increasing the pyrolysis temperature increased the HDPEPO yield, viscosity, and iodine value, whereas the flash point decreased. The higher flash point is needed to ensure safety during production since the polyurethane reaction is exothermic, whereas low iodine value of HDPEPO favours stability of the thermal insulator. The HDPEPO exhibits the property of alcohol due to the presence of n-tetracosanol; however, it cannot completely replace conventional polyol in the production of polyurethane thermal insulator. This is because the HDPEPO contains low hydroxyl value and slightly high acid value. The implication of this is that there would be no enough hydroxyl groups to crosslink with isocyanate. This may be the major reason for a failed HDPEPO thermal insulator formulation when HDPEPO was completely substituted for conventional polyol. However, HDPE-PU thermal insulator formulated from HDPEPO/PEO blend proved to demonstrate better performance as a result of the contribution of hydrophobic hydrocarbon by HDPEPO to the copolymer blend. The percentage of HDPEPO used in the HDPEPO/PEO blend can be increased when higher hydroxyl value of PEO is utilised.

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Statements and Declaration

The authors declare that the research was conducted according to accepted scientific standards and that the reported methods and results are presented without ambiguity. All experiments were conducted using the reagents and procedures described in this manuscript.

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Competing Interest

The authors declared no financial competing interest.

Authors' Contributions

Professor Osemeahon conceived the idea and prepare the methodology, Professor Nkafamiya reviewed the methodology. Dr Akinterinwa proofread the write up, Gidigbi Joshua and Markus carried out the experiment at the laboratory, while Barnabas assisted in recording. All authors contributed to the study conception and design.

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References

[1] Braun, D., Cherdron, H., Rehahn, M., Ritter, H., Voit, B. Polymer synthesis: theory and practice: fundamentals, methods, experiments. *Berlin, Heidelberg: Springer Berlin Heidelberg*, **2001**.
[2] Singh, P., Sharma, V. Integrated plastic waste management: environmental and improved health approaches. *Procedia Environmental Sciences*, **2016**, 35, 692-700.
[3] Pan, D., Su, F., Liu, C., Guo, Z. Research progress for plastic waste management and manufacture of value-added products. *Advanced Composites and Hybrid Materials*, **2020**, 3(4), 443-461.
[4] Law, K.L., Sobkowicz, M.J., Shaver, M.P., Hahn, M.E. Untangling the chemical complexity of plastics to improve life cycle outcomes. *Nature Reviews Materials*, **2024**, 9(9), 657-667.
[5] Gautam, B.P.S., Qureshi, A., Gwasikoti, A., Kumar, V., Gondwal, M. Global scenario of plastic production, consumption, and waste generation and their impacts on environment and human health. In *Advanced Strategies for Biodegradation of Plastic Polymers*, **2024**, 1-34.
[6] George, S. Plastics we cannot live without. In *Plastic Waste and Recycling*, **2020**, 449-466.
[7] Labelnaterials. Retrieved from Label materials. *Packaging Recyclability,* **2025.**
[8] Gill, Y.Q., Jin, J., Song, M. Comparative study of carbon-based nanofillers for improving the properties of HDPE for potential applications in food tray packaging. *Polymers and Polymer Composites*, **2020**, 28(8-9), 562-571.
[9] Moharir, R.V., Kumar, S. Challenges associated with plastic waste disposal and allied microbial routes for its effective degradation: A comprehensive review. *Journal of Cleaner Production*, **2019**, 208, 65-76.
[10] Panda, A.K., Alotaibi, A., Kozhevnikov, I.V., Shiju, N.R. Pyrolysis of plastics to liquid fuel using sulphated zirconium hydroxide catalyst. *Waste and Biomass Valorization*, **2020**, 11(11), 6337-6345.
[11] Gidigbi, J.A., Martins, A., Enyoh, C.E. A problem in disguise: A review paper on generous uses of polyethylene bags (Nylon bags) in Nigeria and its environmental implications. *AIMS Environmental Science*, **2020**, 7(6), 602-610.
[12] OECD. Retrieved from OECD *Global plastics outlook-plastics waste in 2019 by region, polymer and application,* **2022.**
[13] Hopewell, J., Dvorak, R., Kosior, E. Plastics recycling: challenges and opportunities. *Philosophical Transactions of the Royal Society B: Biological Sciences*, **2009**, 364(1526), 2115-2126.
[14] Suvari, F., Gurvardar, H. Revitalizing high-density polyethylene (HDPE) waste: from environmental collection to high-strength hybrid yarns. *Polymer Bulletin*, **2024**, 81(15), 14011-14029.
[15] Chidichimo, F., De Biase, M., Straface, S. Groundwater pollution assessment in landfill areas: Is it only about the leachate? *Waste Management*, **2020**, 102, 655-666.
[16] Orozco, S., Artetxe, M., Lopez, G., Suarez, M., Bilbao, J., Olazar, M. Conversion of HDPE into value products by fast pyrolysis using FCC spent catalysts in a fountain confined conical spouted bed reactor. *ChemSusChem*, **2021**, 14(19), 4291-4300.
[17] Prociak, A., Kurańska, M., Uram, K., and Wójtowicz, M. Bio-polyurethane foams modified with a mixture of bio-polyols of different chemical structures. *Polymers*, **2021**, 13(15), 2469.
[18] Członka, S., Bertino, M.F., Strzelec, K., Strąkowska, A., Masłowski, M. Rigid polyurethane foams reinforced with solid waste generated in leather industry. *Polymer Testing*, **2018**, 69, 225-237.
[19] Osemeahon, S., Dass, P., Aliyu, B., and Fadawa, F. Development of an Emulsion Paint Binder from the Copolymerization of Dimethylol Urea and Hydroxylated Mahogany Seed Oil. *International Journal of Recent Innovations in Academic Research*, **2018**, 2(7), 26-39.
[20] Marwani, M., Trifarizy, M.D. the Pyrolysis of High-Density Polyethylene (Hdpe) and Polypropylene (Pp) Plastic Waste Blend. *Indonesian Journal of Engineering and Science*, **2024**, 5(1), 029-034.
[21] Hendrawati, A., Liandi, A. R., Solehah, M., Setyono, M., Aziz, I., and Siregar, Y. Pyrolysis of PP and HDPE from plastic packaging waste into liquid hydrocarbons using natural zeolite Lampung as a catalyst. *Case Studies in Chemical and Environmental Engineering*, **2023**, 100290.
[22] Prurapark, R., Owjaraen, K., Saengphrom, B., Limthongtip, I., Tongam, N. Effect of temperature on pyrolysis oil using high-density polyethylene and polyethylene terephthalate sources from mobile pyrolysis plant. *Frontiers in Energy Research*, **2020**, 8, 541535.
[23] Rominiyi, O., Akintunde, M., Bello, E., Lajide, L., Ikumapayi, O.M. Performance Evaluation of a Batch: Reactor for Catalytic Depolymerization of Polymeric Waste. In *Localized Energy Transition in the 4th Industrial Revolution, Journal*, **2024**, 86-104.
[24] Sogancioglu, M., Yel, E., Ahmetli, G. Pyrolysis of waste high density polyethylene (HDPE) and low density polyethylene (LDPE) plastics and production of epoxy composites with their pyrolysis chars. *Journal of Cleaner Production*, **2017**, 165, 369-381.

[25] Purwanto, N.S., Chen, Y., Wang, T., Torkelson, J.M. Rapidly synthesized, self-blowing, non-isocyanate polyurethane network foams with reprocessing to bulk networks via hydroxyurethane dynamic chemistry. *Polymer*, **2023**, 272, 125858.
[26] Uebe, J., Lekaviciute, E., Kryzevicius, Z., Zukauskaite, A. Comparison of Antioxidants to Increase the Oxidation Stability of Pyrolysis Oils of Three Plastics Using Iodine Value. *Processes*, **2024**, 12(4), 638.
[27] Dagde, K.K. Extraction of vegetable oil from avocado seeds for production of biodiesel. *Journal of Applied Sciences and Environmental Management*, **2019**, 23(2), 215-221.
[28] Cahyono, M.S., Fenti, U.I. Influence of heating rate and temperature on the yield and properties of pyrolysis oil obtained from waste plastic bag. *Conserve: Journal of Energy and Environmental Studies*, **2017**, 1(1), 1-8.
[29] Gin, A., Hassan, H., Ahmad, M., Hameed, B., Din, A.M. Co-pyrolysis of polyolefin mixtures and oil palm fibre for the production of liquid fuel: kinetics and thermodynamic study. *Biomass Conversion and Biorefinery*, **2024**, 14(5), 6381-6395.
[30] Alawa, B., Chakma, S. Synergism and production of hydrocarbon-rich fuel from mixed-feedstock through co-pyrolysis of LDPE and PP: an assessment of fuel properties, engine performance, and gas emission. *Journal of Analytical and Applied Pyrolysis*, **2022**, 168, 105736.
[31] Singh, S., Dixit, P., Rathore, K.S., Sharma, N. Detrimental Impact of Plastic Outcome on Agriculture: Biggest Threat to Environment. In *Research on Environmental and Human Health Impacts of Plastic Pollution*, **2020**, 122-141.
[32] Rodriguez Lamar, Y., Noboa, J., Torres Miranda, A.S., Almeida Streitwieser, D. Conversion of PP, HDPE and LDPE plastics into liquid fuels and chemical precursors by thermal cracking. *Journal of Polymers and the Environment*, **2021**, 29(12), 3842-3853.
[33] Sharuddin, S.D.A., Abnisa, F., Daud, W.M.A. W., Aroua, M.K. A review on pyrolysis of plastic wastes. *Energy conversion and management*, **2016**, 115, 308-326.
[34] Grigore, M.E. Methods of recycling, properties and applications of recycled thermoplastic polymers. *Recycling*, **2017**, 2(4), 24.
[35] Turku, I., Kärki, T., Rinne, K., Puurtinen, A. Characterization of plastic blends made from mixed plastics waste of different sources. *Waste Management & Research*, **2017**, 35(2), 200-206.
[36] Park, S.H., Kim, S.H. Poly (ethylene terephthalate) recycling for high value added textiles. *Fashion and Textiles*, **2014**, 1(1), 1.
[37] Dutta, A.S. Polyurethane foam Chemistry. *In Recycling of Polyurethane Foams*, **2018**, 17-28.
[38] Babanyaya, A., Osemeahon, S.A., Gidigbi, J.A., Apolmi, G. Copolymerization of dimethylol urea with hydroxylated watermelon seed oil as binder for emulsion paint formulation. *Journal of Basic Physical Research*, **2024**, 12(1), 63-72.
[39] Osemeahon, S.A., Akinterinwa, A., Fasina, E., Andrew, F.P., Shagal, M.H., Kareem, S.A., Esenowo, D. Reduction of polystyrene/polyurethane plastic wastes from the environment into binders for water-resistant emulsion paints. *Heliyon*, **2024**, 10(6).
[40] Osemeahon, S.A., Reuben, U., Emmanuel, E. Development of adhesive from polystyrene waste. *BIOMED Natural and Applied Science*, **2022**, 2(1), 13-24.
[41] Kirpluks, M., Cabulis, U., Ivdre, A., Kuranska, M., Zieleniewska, M., Auguscik, M. Mechanical and thermal properties of high-density rigid polyurethane foams from renewable resources. *Journal of Renewable Materials*, **2016**, 4(1), 86-100.
[42] Ivdre, A., Abolins, A., Sevastyanova, I., Kirpluks, M., Cabulis, U., Merijs-Meri, R. Rigid polyurethane foams with various isocyanate indices based on polyols from rapeseed oil and waste PET. *Polymers*, **2020**, 12(4), 738.
[43] Wakita, K., Kuwabara, H., Furusho, N., Tatebe, C., Sato, K., and Akiyama, H. A Comparative Study of the Hydroxyl and Saponification Values of Polysorbate 60 in International Food Additive Specifications. *American Journal of Analytical Chemistry*, **2014**, 5(3), 199-204
[44] Kairytė, A., Kremensas, A., Balčiūnas, G., Członka, S., Strąkowska, A. Closed cell rigid polyurethane foams based on low functionality polyols: Research of dimensional stability and standardised performance properties. *Materials*, **2020**, 13(6), 1438.
[45] Fu, Y., Qiu, C., Ni, L., Ye, H., Zou, H., Luo, Y., Liang, M. Cell structure control and performance of rigid polyurethane foam with lightweight, good mechanical, thermal insulation and sound insulation. *Construction and Building Materials*, **2024**, 447, 138068.
[46] Amaral, C., Vicente, R., Ferreira, V.M., Silva, T. Polyurethane foams with microencapsulated phase change material: Comparative analysis of thermal conductivity characterization approaches. *Energy and buildings*, **2017**, 153, 392-402.
[47] Suleiman, A.S., Dukhan, N. Long-domain simulation of flow in open-cell mesoporous metal foam and direct comparison to experiment. *Microporous and mesoporous materials*, **2014**, 196, 104-114.
[48] Jabar, J.M., Fakinlede, F.K., Bello, B.E. Synthesis of rigid polyurethane foam using polyols from fractionated locally sourced vegetable oil. *FUW Trends in Science and Technology Journal*, **2018**, 3(2), 384-388.
[49] Vieira, F.R., Gama, N.V., Barros-Timmons, A., Evtuguin, D.V., Pinto, P.C. Development of rigid polyurethane foams based on Kraft lignin polyol obtained by oxyalkylation using propylene carbonate. *ChemEngineering*, **2022**, 6(6), 95.

[50] Septevani, A.A., Evans, D.A., Chaleat, C., Martin, D.J., Annamalai, P.K. A systematic study substituting polyether polyol with palm kernel oil based polyester polyol in rigid polyurethane foam. *Industrial Crops and Products*, **2015**, 66, 16-26.
[51] Kabakci, E., Sayer, G., Suvaci, E., Uysal, O., Güler, İ., Kaya, M. Processing-structure-property relationship in rigid polyurethane foams. *Journal of Applied Polymer Science*, **2017**, 134(21).
[52] Petcu, C., Hegyi, A., Stoian, V., Dragomir, C.S., Ciobanu, A.A., Lăzărescu, A.V., Florean, C. Research on thermal insulation performance and impact on indoor air quality of cellulose-based thermal insulation materials. *Materials*, **2023**, 16(15), 5458.

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